

Skills Summary

Sign and Like-Term Errors in Polynomial Operations

Use this reference while completing your worksheet or when studying.

Skill 1 — Subtracting and Combining Like Terms

Subtracting means adding the opposite: $A - B = A + (-1)B$, so every term of B changes sign.

- Write the sign-flipped line before you combine anything. It is one extra line and it is where the error would have been.
- Terms are *like* only when the variable parts match exactly: $4x^2$ and $-x^2$ combine, $4x^2$ and $-x$ do not.
- Add the coefficients; never touch the exponent. $3x + 5x = 8x$, not $8x^2$.

Example: Simplify $(6x^2 + 2x - 9) - (4x^2 - 5x + 3)$.

Step 1. The minus sign belongs to all three terms of the second polynomial, so rewrite the subtraction as adding the opposite before combining.

Step 2. $6x^2 + 2x - 9 - 4x^2 + 5x - 3$, and grouping like terms gives $2x^2 + 7x - 12$.

Try it: Simplify $(8x^2 - x + 5) - (3x^2 + 4x - 2)$.

check: $5x^2 - 7x + 7$

Skill 2 — Distributing and Expanding

Every term, every time: $-a(b + c) = -ab - ac$, and $(a - b)^2 = a^2 - 2ab + b^2$ — the middle term is not optional.

- Two negatives make a positive: $(-2x)(-7) = +14x$. The most-missed sign on any sheet.
- A square is a product of two identical factors. Write $(2x - 3)(2x - 3)$ out rather than squaring each piece.
- Multiply coefficients, add exponents: $(-3x)(2x^2) = -6x^3$.

Example: Expand $-4x(2x - 3) + 2(x + 6)$.

Step 1. Two separate distributions have to happen before any terms are collected, so handle each parenthesis on its own and keep the signs with the factors that produced them.

Step 2. $-8x^2 + 12x$ from the first, $2x + 12$ from the second, and collecting the x terms gives $-8x^2 + 14x + 12$.

Try it: Expand $(3x - 5)^2$.

check: $9x^2 - 30x + 25$

Skill 3 — Factoring with Signs

For $x^2 + bx + c$: find two numbers whose product is c and whose sum is b .

- c negative $\Rightarrow$ the two numbers have opposite signs, and the one with the larger size takes b 's sign.
- c positive $\Rightarrow$ both numbers share b 's sign.
- Always check: substitute one value, such as $x = 1$, into the original and into your factors. They must agree.

Example: Factor $x^2 - 2x - 15$.

Step 1. The constant is negative, so the two numbers have opposite signs, and the sum being negative means the larger one is the negative.

Step 2. The pair -5 and 3 has product -15 and sum -2 , so the factors are $(x - 5)(x + 3)$.

Try it: Factor $x^2 - 3x - 28$.

(4 + x)(2 - x) :check